

Experiment No : 7

Aim: To study and implement Identity and Access Management (IAM) practices on AWS.

Theory:

AWS Identity and Access Management (IAM)Features

- You can grant other people permission to administer and use resources in your AWS account without having to share your password or access key.
- You can grant different permissions to different people for different resources.
- You can use IAM features to securely provide credentials for applications that run on EC2 instances which provide permissions for your applications to access other AWS resources.
- You can add two-factor authentication to your account and to individual users for extra security.
- You can allow users to use **identity federation** to get temporary access to your AWS account.
- You receive AWS CloudTrail log records that include information about **IAM identities** who made requests for resources in your account.
- You use an **access key** (an access key ID and secret access key) to make programmatic requests to AWS. An Access Key ID and Secret Access Key can only be uniquely generated once and must be regenerated if lost.
- IAM has been validated as being compliant with Payment Card Industry (PCI) Data Security Standard (DSS).
- IAM is *eventually consistent*. IAM achieves high availability by replicating data across multiple servers within Amazon's data centers around the world.
- IAM and AWS Security Token Service (STS) are offered at no additional charge
- You can use IAM tags to add custom attributes to an IAM user or role using a tag key–value pair.
- You can generate and download a credential report that lists all users on your AWS account. The report also shows the status of passwords, access keys, and MFA devices.

Infrastructure Elements

- **Principal**
 - An entity that can make a request for an action or operation on an AWS resource. Users, roles, federated users, and applications are all AWS principals.
 - Your AWS account root user is your *first principal*.
- **Request**
 - When a principal tries to use the AWS Management Console, the AWS API, or the AWS CLI, that principal sends a *request* to AWS.
 - Requests includes the following information:
 - **Actions or operations** – the actions or operations that the principal wants to perform.
 - **Resources** – the AWS resource object upon which the actions or operations are performed.
 - **Principal** – the user, role, federated user, or application that sent the request. Information about the principal includes the policies that are associated with that principal.
 - **Environment data** – information about the IP address, user agent, SSL enabled status, or the time of day.
 - **Resource data** – data related to the resource that is being requested.
- **Authentication**
 - To authenticate from the console as a user, you must sign in with your username and password.
 - To authenticate from the API or AWS CLI, you must provide your access key and secret key.
- **Authorization**
 - AWS uses values from the *request context* to check for policies that apply to the request. It then uses the policies to determine whether to allow or deny there quest.
 - Policies types can be categorized as *permissions policies* or *Permissions boundaries*.
 - *Permissions policies* define the permissions for the object to which they're attached. These include identity-based policies, resource-based policies, and ACLs.
 - *Permissions boundary* is an advanced feature that allows you to use policies to limit the maximum permissions that a principal can have.
 - To provide your users with permissions to access the AWS resources in their own account, you need **identity-based policies**.
 - **Resource-based policies** are for granting cross-account access.
 - Evaluation logic rules for policies:

- By default, **all requests are denied**.
- An *explicit allow* in a permissions policy overrides this default.
- A *permissions boundary* overrides the allow. If there is a permissions boundary that applies, that boundary must allow the request. Otherwise, it is implicitly denied.
- An explicit deny in any policy overrides any allows.
- **Actions or Operations**
 - Operations are defined by a service, and include things that you can do to a resource, such as viewing, creating, editing, and deleting that resource.
- **Resource**
 - An object that exists within a service. The service defines a set of actions that can be performed on each resource.

Users

- **IAM Users**
 - Instead of sharing your root user credentials with others, you can create individual **IAM users** within your account that correspond to users in your organization. IAM users are not separate accounts; they are users within your account.
 - Each user can have its own password for access to the AWS Management Console. You can also create an individual access key for each user so that the user can make programmatic requests to work with resources in your account.
 - By default, a brand new IAM user has **NO permissions** to do anything.
 - Users are global entities.
- **Federated Users**
 - If the users in your organization already have a way to be authenticated, you can federate those user identities into AWS.
- **IAM Groups**
 - An IAM group is a collection of IAM users.
 - You can organize IAM users into IAM groups and attach access control policies to a group.
 - A user can belong to multiple groups.
 - Groups cannot belong to other groups.
 - Groups do not have security credentials, and cannot access web services directly.
- **IAM Role**
 - A role does not have any credentials associated with it.
 - An IAM user can assume a role to temporarily take on different permissions for a specific task. A role can be assigned to a federated user who signs in by using an external identity provider instead of IAM.

- **AWS service role** is a role that a service assumes to perform actions in your account on your behalf. This service role must include all the permissions required for the service to access the AWS resources that it needs.
 - **AWS service role for an EC2 instance** is a special type of service role that a service assumes to launch an EC2 instance that runs your application. This role is assigned to the EC2 instance when it is launched.
 - **AWS service-linked role** is a unique type of service role that is linked directly to an AWS service. Service-linked roles are predefined by the service and include all the permissions that the service requires to call other AWS services on your behalf.
- An instance profile is a container for an IAM role that you can use to pass role information to an EC2 instance when the instance starts.
- Users or groups can have multiple policies attached to them that grant different permissions.

Policies

- Most permission policies are JSON policy documents.
- The IAM console includes *policy summary tables* that describe the access level, resources, and conditions that are allowed or denied for each service in a policy.
- The *policy summary table* includes a list of services. Choose a service there to see the *service summary*.
- This *summary table* includes a list of the actions and associated permissions for the chosen service. You can choose an action from that table to view the action *summary*.
- To assign permissions to federated users, you can create an entity referred to as a **role** and define permissions for the **role**.
- **Identity-Based Policies**
 - Permissions policies that you attach to a principal or identity.
 - **Managed policies** are standalone policies that you can attach to multiple users, groups, and roles in your AWS account.
 - **Inline policies** are policies that you create and manage and that are embedded directly into a single user, group, or role.
- **Resource-based Policies**
 - Permissions policies that you attach to a resource such as an Amazon S3 bucket.
 - Resource-based policies are only inline policies.
 - **Trust policies** – resource-based policies that are attached to a role and define which principals can assume the role.

Procedure :

Step 1: Login to the AWS Management Console using the root account and open the IAM service

Step 2: Create IAM User (Console Access , In IAM, click Users from the left menu, select Create user, enter the user name (your name & roll no.), choose Provide user access to the AWS Management Console, set a Custom password, enable Users must create a new password at next sign-in, and click Next.

us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/create

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Sanika Joshi

Step 3
Review and create

Step 4
Retrieve password

User name

SS

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # % ^ & * () _ + - (hyphen) = [] { } | ' "

☐ Show password

☒ **Users must create a new password at next sign-in - Recommended**
Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

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Step 3 : Assign Permissions (Limited Access) In the permissions section, select Attach policies directly. Search and select **AmazonS3ReadOnlyAccess**.

us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/create

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Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1446) [Create policy](#)

Choose one or more policies to attach to your new user.

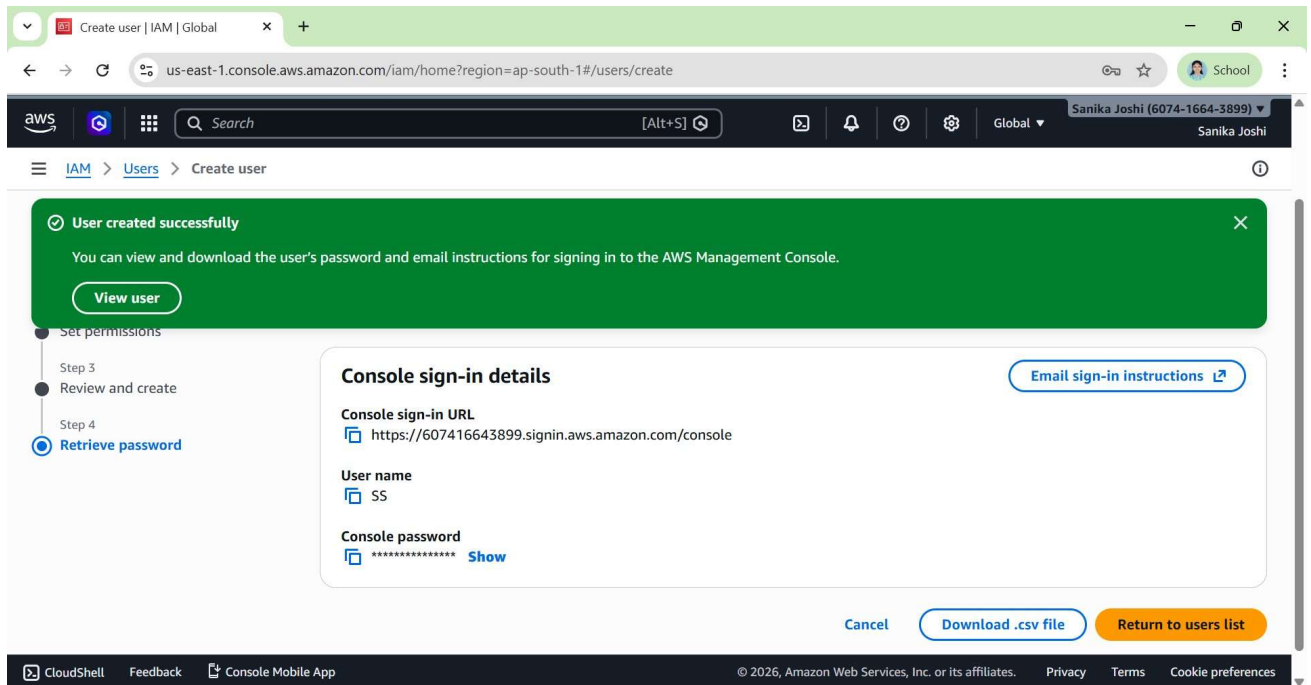
Filter by Type

amazonS3 1 match

☒	Policy name	Type	Attached entities
☒	AmazonS3ReadOnlyAccess	AWS managed	0

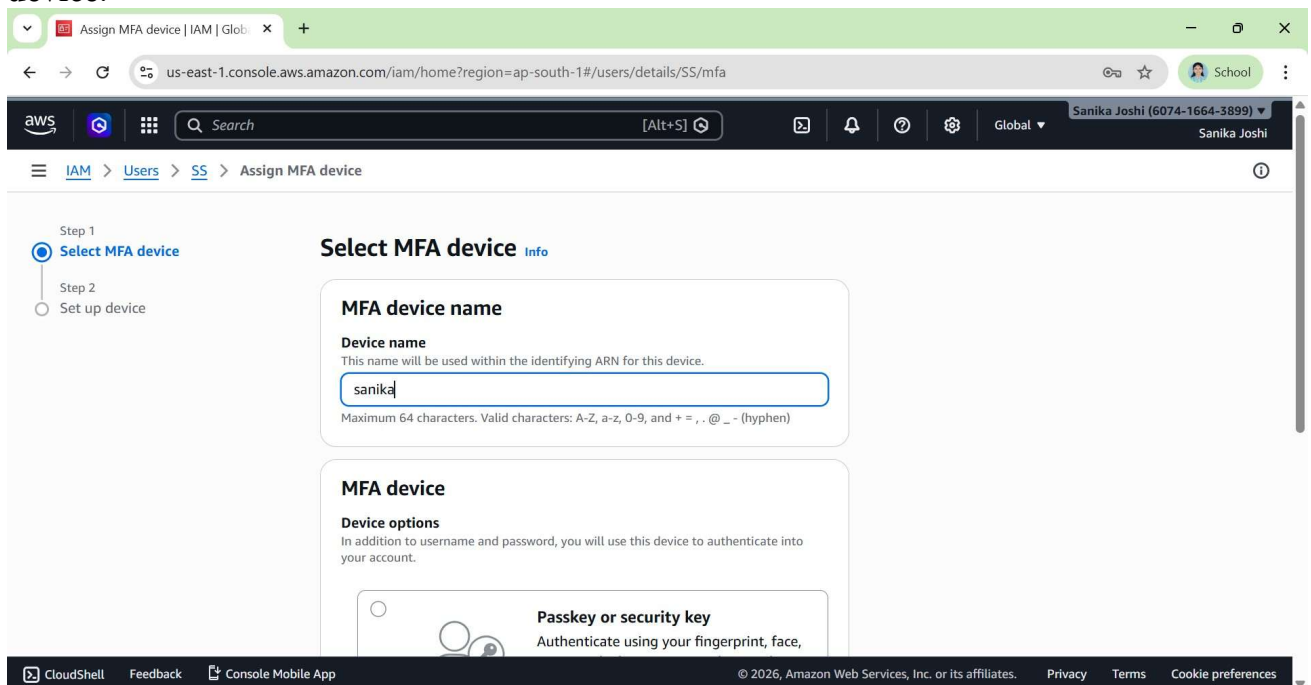
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Step 4 : Click **Next**. Click **Create user**. Download the **.csv file** or copy the **Console sign-in URL**, username, and password.

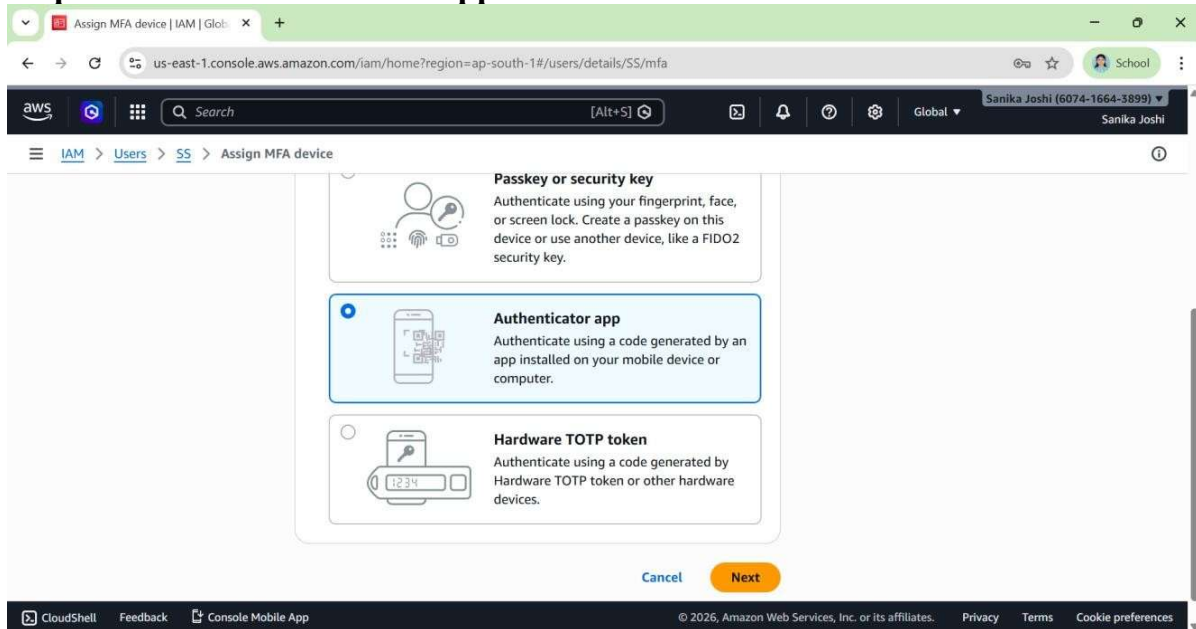


Step 5: Open the IAM user sign-in URL in a new tab, log in with the IAM credentials, change the password, then verify **EC2 access is denied** and **S3 access is available in read-only mode**.

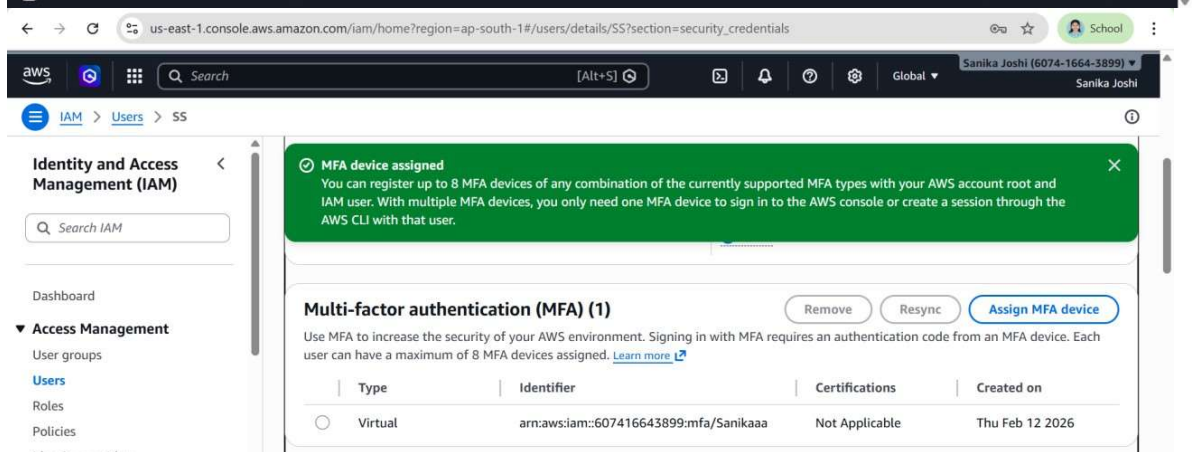
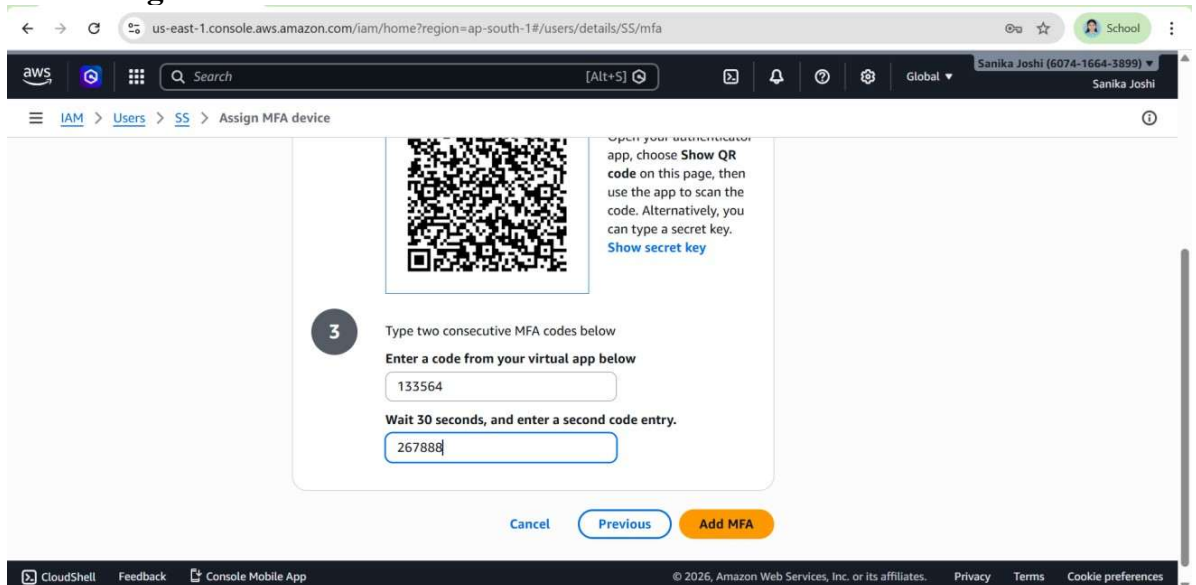
Step 6: Log out from the IAM user, switch back to the root/admin account, open IAM → Users → student-user1 → Security credentials, and under **MFA** click **Assign MFA device**.



Step 7 : Select Authenticator app and click Next.



Step 8 : Install Google Authenticator or Microsoft Authenticator on mobile. Scan the QR code using the authenticator app. Enter two consecutive OTP codes and click Assign MFA.



Step 9 :Open the IAM user sign-in URL again. Login using the IAM user credentials.
Enter the MFA OTP code from the authenticator app and verify successful login.

Step 10 : Create IAM group

Logout and return to the root/admin account. In IAM, click on **User groups** from the left menu.

Click on **Create group**. Enter the group name as (Enter unique name).

In permissions, attach the policies: **AmazonEC2FullAccess** **AmazonS3FullAccess**
Add users to the group (example: student-user2, student-user3). Click **Create group**.

Step 11 :

In IAM, click on **Users** and select **Create user**. Create 2 or 3 more users (example: student-user2, student-user3).

Enable console access and set passwords. In permissions, select **Add user to group**.
Select the group (Your Group). Click **Create user**.

Step 12 :Login as student-user2 using the IAM sign-in URL.

Open EC2 and verify the user can launch an instance.

Open S3 and verify the user can create a bucket. Delete the created bucket after testing to avoid unnecessary resources.

Step 13 : Logout and return to the root/admin account.

In IAM, click on **Roles** from the left menu. Click **Create role**.

Select **AWS service** as the trusted entity. Select **EC2** as the use case and click **Next**.
Attach the policy **AmazonS3ReadOnlyAccess**.

Click **Next** and enter the role name as **EC2-S3ReadOnly-Role**. Click **Create role**.

Step 14 : In IAM, delete the created users (student-user1, student-user2, student-user3).

Delete the IAM group **CloudStudentsGroup**.

Delete the IAM role **EC2-S3ReadOnly-Role**. Verify the IAM dashboard after cleanup.

Chart Analysis:

Below are **3 sample charts similar to those seen in a Mutual Fund Consolidated Account Statement (CAS)**. The values are realistic but are only for demonstration purposes.

1. Asset Allocation Chart (Portfolio Distribution)

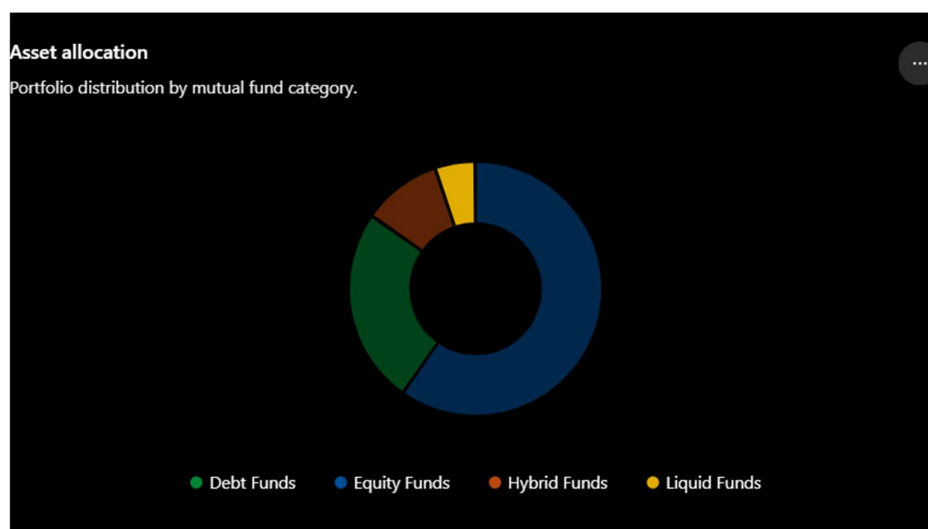


Fig. No. 4.1: Asset Allocation graph

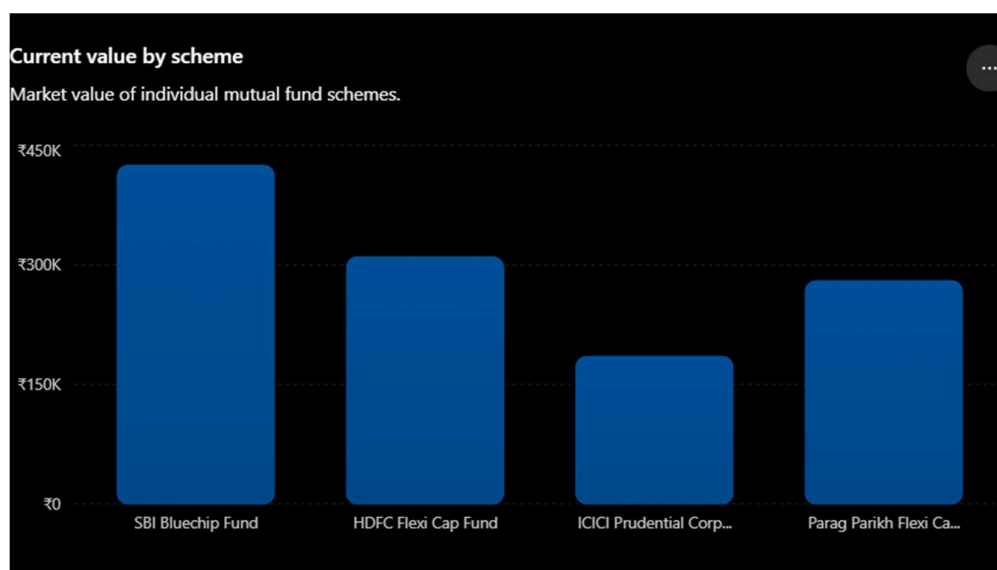
This chart shows how the total investment of ₹12,00,000 is distributed across different fund categories.

Summary

Fund Type	Amount (₹)	Allocation
Equity Funds	7,20,000	60%
Debt Funds	3,00,000	25%
Hybrid Funds	1,20,000	10%
Liquid Funds	60,000	5%
Total	12,00,000	100%

Table no. 4.1: Asset Allocation Chart

2. Scheme-wise Current Market Value

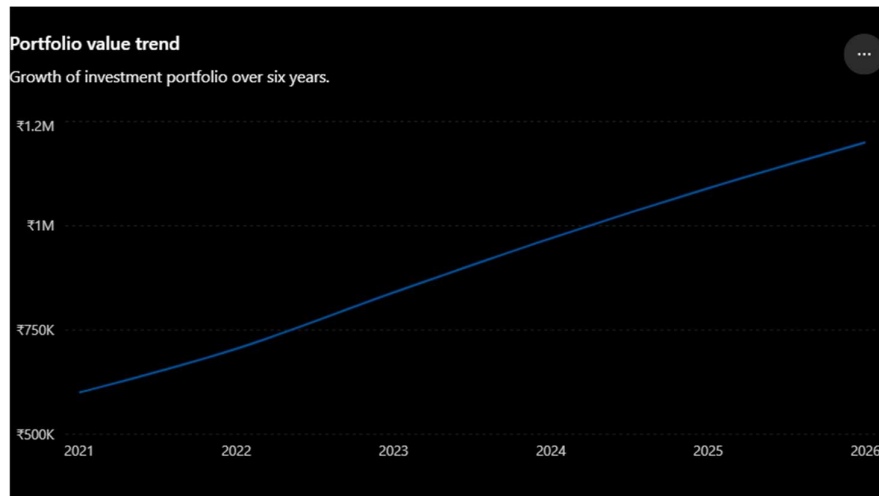


This resembles the scheme-wise summary generally shown in CAS statements.

Scheme Details

Scheme Name	Current Value (₹)
SBI Bluechip Fund	4,25,000
HDFC Flexi Cap Fund	3,10,000
ICICI Prudential Corporate Bond Fund	1,85,000
Parag Parikh Flexi Cap Fund	2,80,000
Total	12,00,000

3. Growth of Portfolio Value Over Time



A CAS performance report often contains portfolio growth information.

Portfolio Performance

Year	Portfolio Value (₹)
2021	6,00,000
2022	7,05,000
2023	8,40,000
2024	9,70,000
2025	10,90,000
2026	12,00,000

Other chart types commonly found in Mutual Fund CAS reports

1. AMC-wise Exposure

- SBI Mutual Fund – 35%
- HDFC Mutual Fund – 25%
- ICICI Prudential Mutual Fund – 20%
- Parag Parikh Mutual Fund – 20%

2. Capital Gain Summary

- Equity LTCG
- Equity STCG
- Debt LTCG
- Debt STCG

3. Transaction Trend

- Monthly SIP Amounts

- Additional Purchases
- Redemptions

4. XIRR / CAGR Performance Chart

- 1 Year Return
- 3 Year Return
- 5 Year Return
- Since Inception Return